

**Spring 2025**

GRADE 8  
**MATHEMATICS**

PTM0822\_3,1

1. Solve the equation  $4x^2 = 256$ . What are the possible values of  $x$ ?

Select **all** the correct answers.

**A.**  $-8$

**B.**  $0$

**C.**  $8$

**D.**  $4$

**E.**  $16$

PTM0824\_P

2. The table represents a function.

| $x$ | $y$ |
|-----|-----|
| 1   | 2   |
| 2   | 3   |
| 3   | 4   |
|     | 5   |

What values of  $x$  can be added to the table to remain a function?

Select **all** the correct answers.

**A.**  $x = 0$

**B.**  $x = 1$

**C.**  $x = 2$

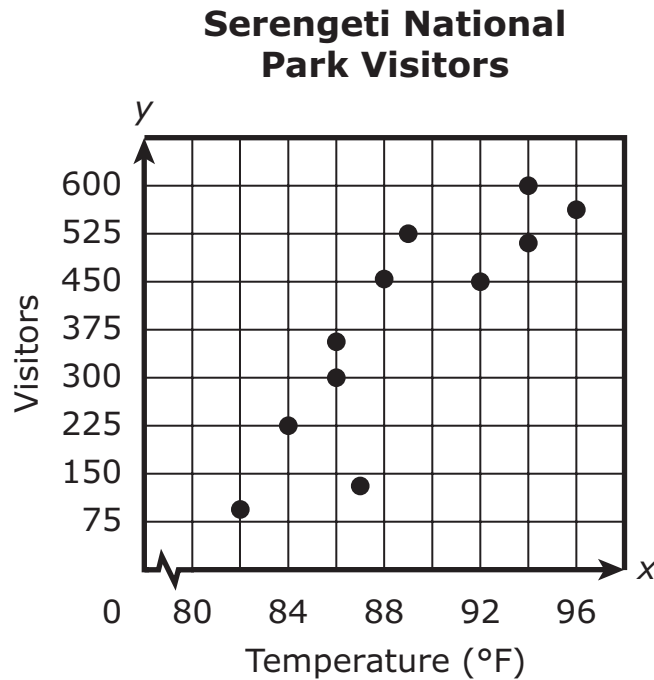
**D.**  $x = 3$

**E.**  $x = 4$

**F.**  $x = 5$

PTM0819\_1

3. The scatter plot shows the relationship between the temperature, in degrees Fahrenheit, and the number of visitors to Serengeti National Park in Tanzania.



Which of these is the best description of the type of relationship between the temperature and the number of visitors to Serengeti National Park?

- A. Positive linear association
- B. Negative linear association
- C. Nonlinear association
- D. No association

PTM0802\_8

4. In 2021, Washington, D.C., had the highest population density in the United States, with approximately  $11 \times 10^3$  people per square mile. New Jersey had the second highest population density, with approximately  $13 \times 10^2$  people per square mile.

To the nearest whole number, how many times the population density of New Jersey was the population density of Washington, D.C.?

Enter your answer in the space.

\_\_\_\_\_

PTM0825\_P

5. Which sets of ordered pairs represent a function?

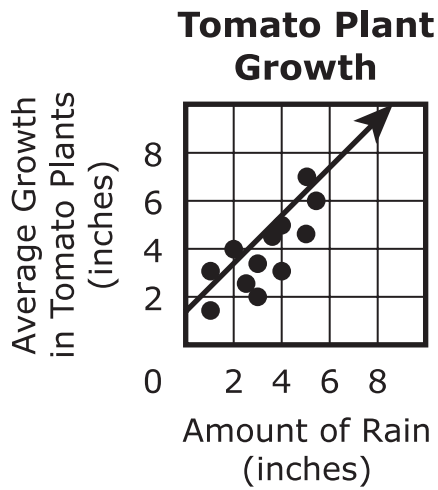
Select **all** the correct answers.

- A.** Set A: (2, 3), (4, 5), (6, 7), (8, 9)
- B.** Set B: (2, 3), (2, 4), (3, 5), (4, 6)
- C.** Set C: (1, 2), (2, 4), (3, 6), (4, 8)
- D.** Set D: (2, 2), (2, 3), (2, 4), (2, 5)
- E.** Set E: (1, 2), (2, 2), (3, 2), (4, 2)

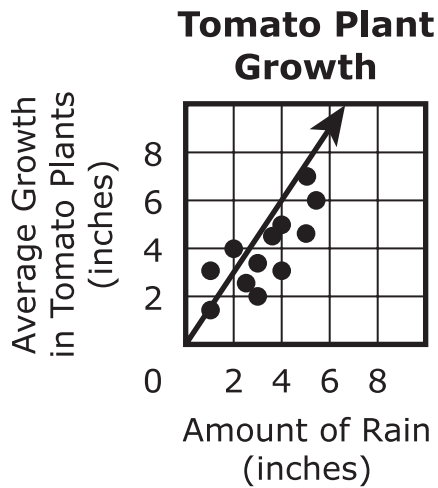
PTM0820\_4

6. Which graph shows the line of best fit for the scatter plot?

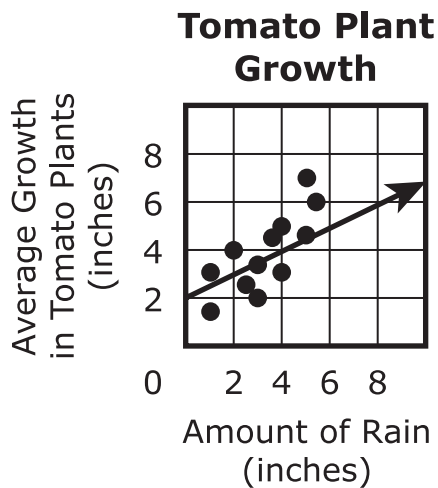
A.



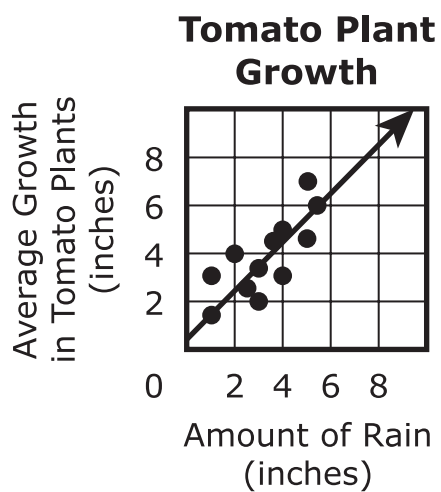
B.



C.



D.



PTM0808\_6

7. Solve for  $x$ .

$$7x - 4 = 2(15 - x) + 20$$

Enter your answer in the space.

\_\_\_\_\_

M18\_D925\_2

8. Which expression represents the solution to the equation shown?

$$x^3 = 2$$

A.  $\sqrt[2]{3}$

B.  $\sqrt[3]{2}$

C.  $2\sqrt{3}$

D.  $3\sqrt{2}$

PTM0809\_3

9. A system of linear equations is shown.

$$y = 2x - 1$$

$$y = -3x + 4$$

Which statement describes the solution to the system of equations?

- A. The solution is a point that is on the line  $y = 2x - 1$  but not on the line  $y = -3x + 4$ .
- B. The solution is a point that is on the line  $y = -3x + 4$  but not on the line  $y = 2x - 1$ .
- C. The solution is the point where the two lines intersect.
- D. The solution is a point that is not on either line.

M18\_B765\_3

10. Which expression has a value between 11 and 12?

A.  $\sqrt{115}$

B.  $\sqrt{120}$

C.  $\sqrt{130}$

D.  $\sqrt{145}$

PTM0801\_4,1

- 11.** Which expressions are equivalent to the expression  $\left(\frac{4^3}{4^{-2}}\right)^2$ ?

Select **all** the correct answers.

**A.**  $\frac{4^6}{4^{-4}}$

**B.**  $\frac{4^5}{4^0}$

**C.**  $\frac{4^1}{4^{-4}}$

**D.**  $(4^5)^2$

**E.**  $(4^1)^2$

8M20\_002\_2

- 12.** What value of  $x$  makes the equation  $(17^{-8})(17^{15}) = \frac{1}{17^x}$  true?

**A.**  $-23$

**B.**  $-7$

**C.**  $7$

**D.**  $23$

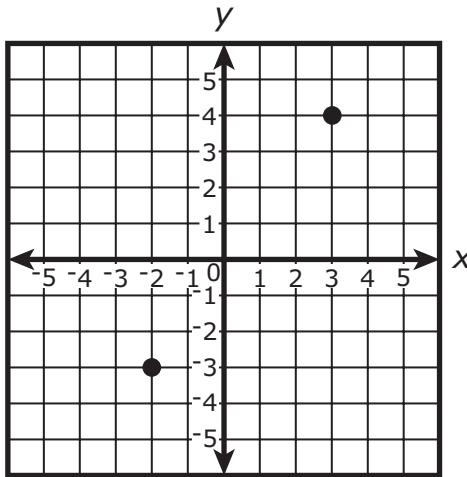
PTM0805

- 13.** Lydia is riding her bike at a constant rate of speed. She travels 3 miles in 18 minutes. Bryce is riding his bike at a constant rate of speed. The equation  $y = 5x$  represents the number of minutes,  $y$ , that it takes Bryce to ride  $x$  miles.
- Show or explain how to determine Lydia's rate of speed in minutes per mile.
  - Show or explain how to determine Bryce's rate of speed in minutes per mile.
  - Who is traveling faster? How much faster?

Enter your answer in the space provided. Show all your work to support your answer.

PTM0816

- 14.** Two points are shown on the coordinate plane.



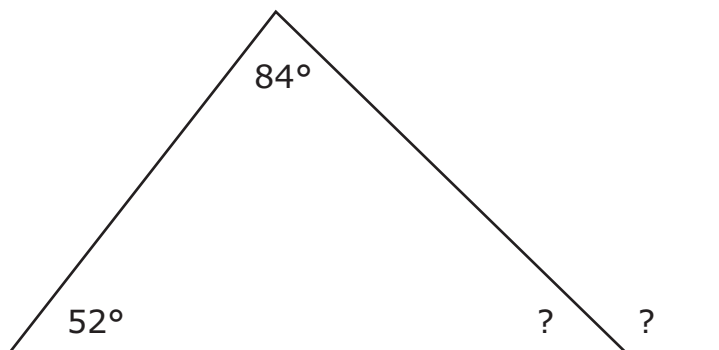
Use the points on the coordinate plane.

- Why can the Pythagorean theorem be used to find the distance between the two points in the coordinate plane?
- Write an equation that can be used to find the distance between the two points in the coordinate plane.
- What is the distance between the two points in the coordinate plane?

Enter your answer in the space provided. Show all your work to support your answer.

PTM0814

- 15.** A diagram of a triangle is shown. The measure of an interior angle and the measure of an exterior angle are missing.



Use the diagram.

- How can you use the two given interior angle measures to find the missing exterior angle measure?
- How can you use the missing interior angle measure to find the missing exterior angle measure?
- What is the missing exterior angle measure?

Enter your answer in the space provided. Show all your work to support your answer.

- 16.** Function  $K$  is represented by  $y = 2x - 7$ .

Function  $L$  is represented by the table.

| $x$ | $y$ |
|-----|-----|
| -2  | 11  |
| -1  | 8   |
| 0   | 5   |
| 1   | 2   |
| 2   | -1  |

Which function has the greater rate of change?

- A.** Function  $K$ , because the  $y$ -intercept of function  $K$  is greater than the  $y$ -intercept of function  $L$
- B.** Function  $L$ , because the slope of function  $L$  is greater than the slope of function  $K$
- C.** Function  $L$ , because the  $y$ -intercept of function  $L$  is greater than the  $y$ -intercept of function  $K$
- D.** Function  $K$ , because the slope of function  $K$  is greater than the slope of function  $L$

PTM0821\_:

**17.** This question has two parts.

Maria, Alex, David, and Jackie each threw a baseball and kicked a football into the air.

**Part A**

Maria threw the baseball 75 feet. Alex threw the baseball a distance 10.5 feet shorter than the distance Maria threw the baseball. David threw the baseball 12.75 feet farther than Alex did. Jackie threw the baseball 1.6 times as far as David did.

- Write an expression to determine the distance Jackie threw the baseball.
- How far did Jackie throw the baseball?
- Show your work.

Enter your answer in the space provided. Show all your work to support your answer.

**Part B**

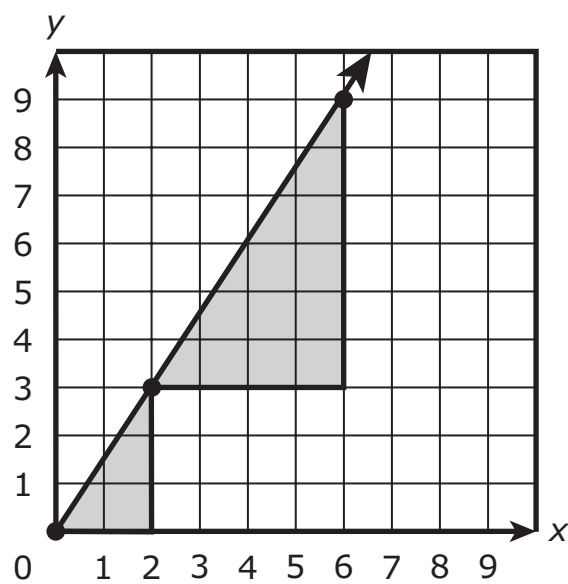
Maria kicked the football 8 feet into the air. Alex kicked the football 1.5 feet higher than Maria did. David kicked the football 2.25 feet lower than Alex did. Jackie kicked the football 1.2 times as high as David did.

- Write an expression to determine the height Jackie kicked the football into the air.
- How high did Jackie kick the football into the air?
- Show your work.

Enter your answer in the space provided. Show all your work to support your answer.

PTM0807\_1

- 18.** Two similar right triangles are shown.

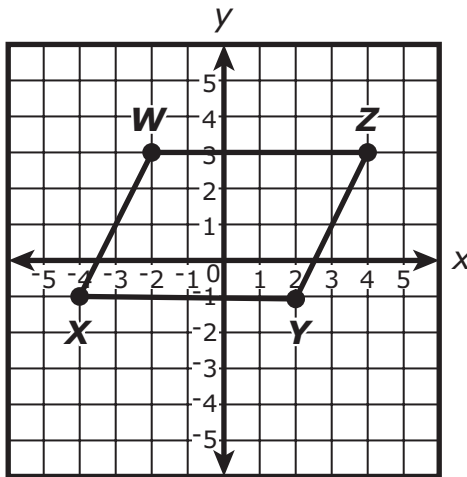


Which statement describes the relationship between the hypotenuses of the similar triangles and the slope of the line?

- A.** The ratio of rise to run for the hypotenuses of both triangles is equivalent to  $\frac{3}{2}$ , which means the slopes between the pairs of points on the line are the same.
- B.** The ratio of rise to run for the hypotenuses of both triangles is equivalent to  $\frac{2}{3}$ , which means the slopes between the pairs of points on the line are the same.
- C.** The ratio of rise to run for the hypotenuse of the bottom triangle is  $\frac{3}{2}$ . The ratio of rise to run for the hypotenuse of the top triangle is  $\frac{6}{4}$ . The ratios are not equivalent, which means the slope of a line can be different between pairs of points on the same line.
- D.** The ratio of rise to run for the hypotenuse of the bottom triangle is  $\frac{2}{3}$ . The ratio of rise to run for the hypotenuse of the top triangle is  $\frac{4}{6}$ . The ratios are not equivalent, which means the slope of a line can be different between pairs of points on the same line.

PTM0827\_1,4,3

19. Parallelogram  $WXYZ$  is shown on the coordinate plane.



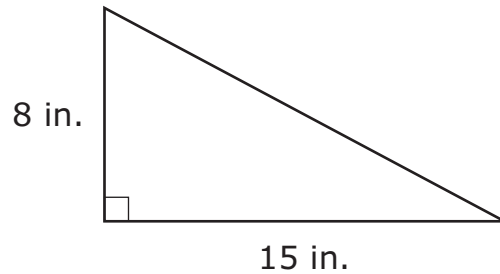
Parallelogram  $WXYZ$  undergoes a translation 2 units to the right and 1 unit up and a rotation 90 degrees clockwise about the origin to produce parallelogram  $W'X'Y'Z'$ . Which statements are true?

Select **all** the correct answers.

- A.** Angle  $X'$  will measure  $75^\circ$ .
- B.** Angle  $X'$  will measure  $165^\circ$ .
- C.** Line segment  $X'Y'$  will be parallel to line segment  $W'Z'$ .
- D.** Line segment  $W'X'$  is the same length as line segment  $WX$ .
- E.** Line segment  $W'Z'$  will be 3 units longer than line segment  $WZ$ .

PTM0815\_17

- 20.** Sam is making a flag for the New York Dance Parade. The flag is a right triangle with one leg 8 inches long and the other leg 15 inches long.



What is the length of the hypotenuse of the triangle in inches?

Enter your answer in the space.

\_\_\_\_\_

M18\_A619\_P

- 21.** A linear function includes the points  $(1, 3)$  and  $(-1, -5)$ .

What are the slope and  $y$ -intercept of the function?

Select **all** the correct answers.

- A.** Slope =  $-4$
- B.** Slope =  $-1$
- C.** Slope =  $4$
- D.**  $(0, -4)$
- E.**  $(0, -1)$
- F.**  $(0, 4)$

PTM0826\_P\_2,4:3

**22.** This question has two parts.

**Part A**

Which equation defines a linear function and why?

Select **all** the correct answers.

**A.**  $y = 3x(x + 2)$

**B.**  $4y + 7 = 3x - 1$

**C.**  $x = -3$

**D.** Its graph contains only points on a straight, non-vertical line.

**E.** Its graph contains some points on a straight, non-vertical line.

**F.** Its graph contains points not on a straight, non-vertical line.

**Part B**

Consider the given equation.

$$y = 10.7x^3 - 4$$

Which value in the equation indicates that the equation defines a nonlinear function?

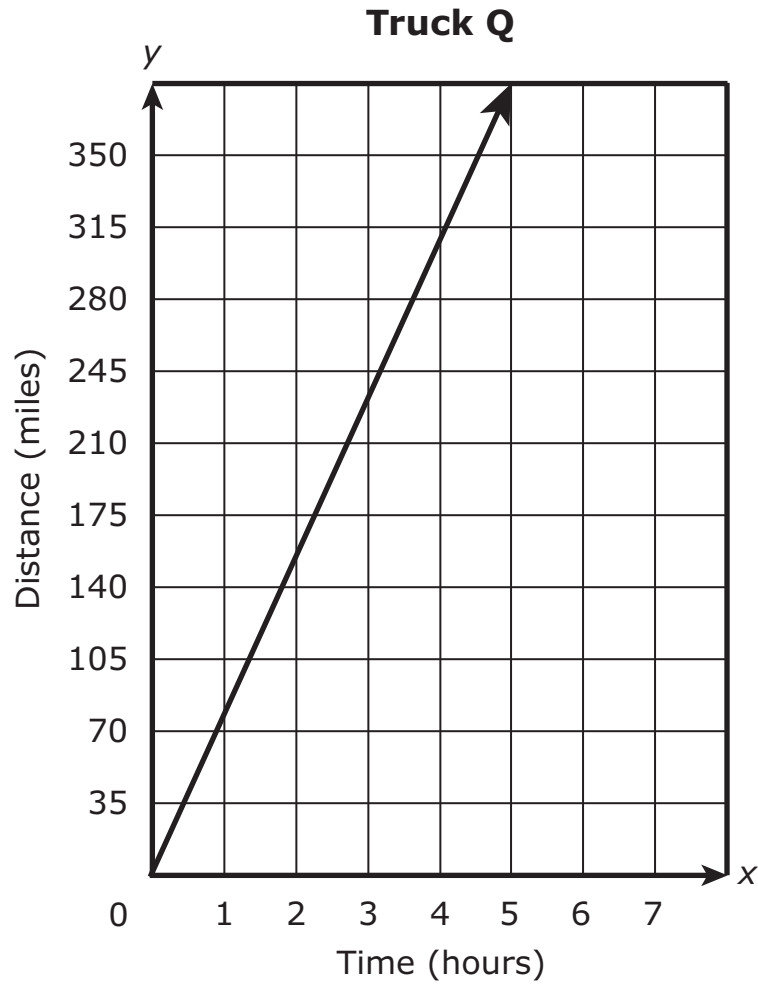
Enter your answer in the space.

\_\_\_\_\_

PTM0804\_3

- 23.** Truck P travels 110 miles in 2 hours, driving at a constant rate.

The graph shows the relationship between the time and the distance that Truck Q travels.



Which statement accurately describes the two proportional relationships?

- A.** Truck P travels 55 miles per hour. Truck Q travels 55 miles per hour.  
Truck P is traveling at the same rate as Truck Q.
- B.** Truck P travels 70 miles per hour. Truck Q travels 55 miles per hour.  
Truck P is traveling at a greater rate than Truck Q.
- C.** Truck P travels 55 miles per hour. Truck Q travels 70 miles per hour.  
Truck Q is traveling at a greater rate than Truck P.
- D.** Truck P travels 70 miles per hour. Truck Q travels 70 miles per hour.  
Truck P is traveling at the same rate as Truck Q.

PTM0823\_:

**24.** This question has two parts.

A system of equations is shown.

$$\begin{cases} y = 2x + 3 \\ y = -x + 1 \end{cases}$$

**Part A**

- Identify the point of intersection when both equations are graphed on the same coordinate plane.
- Explain using words or symbols why the point of intersection represents the solution to the system of equations.

Enter your answer in the space provided.

**Part B**

- Solve the same system of equations algebraically.
- Does your solution match the point of intersection in Part A? Explain why or why not.

Enter your answer in the space provided.

PTM0806

- 25.** There is a proportional relationship between the number of nights,  $x$ , visitors stay at a hotel and the cost in dollars,  $y$ . The equation  $y = 175x$  represents the cost in dollars,  $y$ , to stay at the Atlas Hotel for  $x$  nights. The table shows how much it costs in dollars,  $y$ , to stay at the Emerald Hotel for  $x$  nights.

**Emerald Hotel Costs**

| Number of Nights, $x$ | Cost (dollars), $y$ |
|-----------------------|---------------------|
| 3                     | 540                 |
| 5                     | 900                 |

- Show or explain how to determine the rate in dollars per night at the Atlas Hotel.
- Show or explain how to determine the rate in dollars per night at the Emerald Hotel.
- Which hotel charges less per night? How much less?

Enter your answer in the space provided. Show all your work to support your answer.

8M23\_026\_4

- 26.** Which equation has infinitely many solutions?

**A.**  $-x = x$

**B.**  $2x + 1 = 2x - 1$

**C.**  $x + 6 = 5x + 30$

**D.**  $3(x + 2) = 3x + 6$



## Illinois MATH Assessment

### Practice Item Answer Key

#### Grade 8 - Paper, Screen Reader, and Non-Screen Reader

The following pages include the answer key for all machine-scored items, followed by a sample response for the hand-scored item.

- The rubrics show sample student responses. Student responses other than that shown in the rubric may earn full or partial credit.
- Which responses to hand-scored items receive full or partial credit will be confirmed during range-finding (reviewing sets of real student work)
- If students make a computation error, they can still earn points for reasoning or modeling.

| Item Number | Answer Key             |
|-------------|------------------------|
| 1.          | C,A                    |
| 2.          | E,F,A                  |
| 3.          | A                      |
| 4.          | Student response is 8. |
| 5.          | E, C, A                |
| 6.          | D                      |
| 7.          | Student response is 6. |
| 8.          | B                      |
| 9.          | C                      |
| 10.         | C                      |
| 11.         | D, A                   |
| 12.         | B                      |
| 13.         | See Rubric             |
| 14.         | See Rubric             |
| 15.         | See Rubric             |
| 16.         | D                      |
| 17.         | See Rubric             |



|     |                                           |
|-----|-------------------------------------------|
| 18. | <b>A</b>                                  |
| 19. | <b>A, D, C</b>                            |
| 20. | <b>Student response is 17.</b>            |
| 21. | <b>C, E</b>                               |
| 22. | <b>Part A: B See Rubric<br/>Part B: 3</b> |
| 23. | <b>C</b>                                  |
| 24. | <b>See Rubric</b>                         |
| 25. | <b>See Rubric</b>                         |
| 26. | <b>D</b>                                  |



| #13 Rubric |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Score      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3          | <p>Student response includes the following elements.</p> <ul style="list-style-type: none"><li>• <b>Computation component</b> = 1 point: Identify how much faster Bryce is traveling</li><li>• <b>Modeling component</b> = 1 point: Explain how to determine Lydia's rate of speed</li><li>• <b>Modeling component</b> = 1 point: Explain how to determine Bryce's rate of speed</li></ul> <p>Sample Student Response:</p> <p>Lydia travels 3 miles ÷ 18 minutes, or 1 mile in 6 minutes. The slope of the equation is 5, so Bryce travels 1 mile in 5 minutes. Bryce is riding his bike at a faster rate. Bryce is riding his bike <math>6 - 5 = 1</math> minute faster per mile.</p> <p>Or other valid approaches are acceptable.</p> |
| 2          | Student response includes 2 of the 3 elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1          | Student response includes 1 of the 3 elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 0          | Student response is incorrect or irrelevant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| #14 Rubric |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Score      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3          | <p>Student response includes the following elements.</p> <ul style="list-style-type: none"><li>• <b>Reasoning component</b> = 1 point: Explains why the Pythagorean Theorem can be used to find the distance between the two points in the coordinate plane</li><li>• <b>Modeling component</b> = 1 point: Correct equation that can be used to find the distance between the two points in the coordinate plane</li><li>• <b>Computation component</b> = 1 point: Correct distance between the two points in the coordinate plane</li></ul> |



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <p>Sample Student Response:</p> <p>The Pythagorean Theorem can be used to find the distance between the two points in the coordinate plane because a right triangle can be formed with the two given points as the hypotenuse.</p> <p>A right triangle formed by the two given points has leg lengths of 5 units and 7 units. So, the equation is <math>5^2 + 7^2 = c^2</math>, where <math>c</math> is the distance between two points in the coordinate plane.</p> <p>Solving the equation <math>5^2 + 7^2 = c^2</math> for <math>c</math> gives <math>c = \sqrt{74}</math>, so the distance between two points in the coordinate plane is <math>\sqrt{74}</math> units.</p> <p>Or other valid approaches are acceptable.</p> |
| 2 | Student response includes 2 of the 3 elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1 | Student response includes 1 of the 3 elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## #15 Rubric

| Score | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3     | <p>Student response includes the following elements.</p> <ul style="list-style-type: none"><li>• <b>Reasoning component</b> = 1 point: Explains how to use the two given interior angle measures to find the measure of the missing exterior angle</li><li>• <b>Reasoning component</b> = 1 point: Explains how to use the missing interior angle measure to find the measure of the missing exterior angle</li><li>• <b>Computation component</b> = 1 point: Correct measure of the missing exterior angle</li></ul> <p>Sample Student Response:</p> <p>The sum of the two given interior angle measures is equal to the measure of the missing exterior angle. So, the measure of the missing exterior angle is <math>84^\circ + 52^\circ</math>, or <math>136^\circ</math>.</p> <p>The sum of the angle measures in a triangle is <math>180^\circ</math>. So, the missing interior angle measure is <math>180^\circ - (84^\circ + 52^\circ)</math>, or <math>44^\circ</math>. The measure of the missing exterior angle and the measure of the missing interior angle have a sum of <math>180^\circ</math> since they form a straight line. So, the measure of the missing exterior angle is <math>180^\circ - 44^\circ</math>, or <math>136^\circ</math>.</p> <p>Or other valid approaches are acceptable.</p> |



|   |                                                |
|---|------------------------------------------------|
| 1 | Student response includes 1 of the 3 elements. |
| 0 | Student response is incorrect or irrelevant.   |

## #17 Rubric

### 3 Point Constructed Response Rubric – Part A

| Score | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3     | <p>Student response includes the following elements.</p> <ul style="list-style-type: none"><li>• <b>Modeling component</b> = 1 point: Correct expression to determine the distance Javier threw his baseball</li><li>• <b>Computation component</b> = 1 point: Correct distance Javier threw his baseball</li><li>• <b>Modeling component</b> = 1 point: Correct work shown</li></ul> <p>Sample Student Response:</p> <p>Maria threw the baseball 75 feet. Alex threw the baseball a distance 10.5 feet shorter than the distance Maria threw the baseball. David threw the baseball 12.75 feet farther than Alex did. Jackie threw the baseball 1.6 times as far as David did. So, the expression <math>[(75 - 10.5) + 12.75] \times 1.6</math> can be used to determine the distance Jackie threw the baseball.</p> <p>Jackie threw the baseball <math>[(75 - 10.5) + 12.75] \times 1.6 = (64.5 + 12.75) \times 1.6 = 77.25 \times 1.6 = 123.6</math> feet. Or other valid approaches are acceptable.</p> |
| 2     | Student response includes 2 of the 3 elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1     | Student response includes 1 of the 3 elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0     | Student response is incorrect or irrelevant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

### 3 Point Constructed Response Rubric – Part B

| Score | Description                                                                                                                                                                                                  |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3     | <p>Student response includes the following elements.</p> <ul style="list-style-type: none"><li>• <b>Modeling component</b> = 1 point: Correct expression to determine the height Javier kicked his</li></ul> |



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <p>football into the air</p> <ul style="list-style-type: none"><li>• <b>Computation component</b> = 1 point: Correct distance Javier kicked his football into the air</li><li>• <b>Modeling component</b> = 1 point: Correct work shown</li></ul> <p>Sample Student Response: Maria kicked the football 8 feet into the air. Alex kicked the football 1.5 feet higher than Maria did. David kicked the football 2.25 feet lower than Alex did. Jackie kicked the football 1.2 times as high as David did. So, the expression <math>[(8 + 1.5) - 2.25] \times 1.2</math> can be used to determine the height Jackie kicked the football into the air.</p> <p>Jackie kicked the football <math>[(8 + 1.5) - 2.25] \times 1.2 = (9.5 - 2.25) \times 1.2 = 7.25 \times 1.2 = 8.7</math> feet into the air.</p> <p>Or other valid approaches are acceptable.</p> |
| 2 | Student response includes 2 of the 3 elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1 | Student response includes 1 of the 3 elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0 | Student response is incorrect or irrelevant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## #22 Rubric

### 1 point Rubric: Part A

| Score | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <p>Student response is A. "<math>4x+7=3x-1</math>" and D. "only points on a straight non-vertical line"</p> <p>Rationale:</p> <p>The student understood that of the three given equations, <math>4x+7=3x-1</math> is the only equation that can be rewritten in the form of <math>y=mx+b</math>.</p> <p>The student understood that linear functions can only be defined by equations that have graphs containing only points on a non-vertical straight line.</p> |

### 1 point Rubric: Part B

| Score | Description                                                                                                                                                                          |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <p>Student response is 3.</p> <p>Rationale: The student understood that the term with the variable of "x" should have an exponent with a value of 1 to define a linear function.</p> |



## #24 Rubric

### 2 Point Constructed Response Rubric – Part A

| Score | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2     | <p>Student response includes the following elements.</p> <ul style="list-style-type: none"><li>• <b>Reasoning/Modeling component</b> = 1 point: Correctly describes that the point of intersection represents the solution because the point lies on both lines.</li><li>• <b>Computation component</b> = 1 point: Correct point of intersection as <math>(-2/3, 5/3)</math></li></ul> <p>Sample Student Response:</p> <p>The two lines intersect at <math>(-2/3, 5/3)</math>. The intersection point is where the two lines meet on the graph. This point satisfies both equations because it lies on both lines. If you substitute the x-coordinate of the intersection point into both equations, you get the same y-coordinate for both. Or other valid approaches are acceptable.</p> |
| 1     | Student response includes 1 of the 2 elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0     | Student response is incorrect or irrelevant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### 2 Point Constructed Response Rubric – Part B

| Score | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2     | <p>Student response includes the following elements.</p> <ul style="list-style-type: none"><li>• <b>Reasoning/Modeling component</b> = 1 point: Correctly explains why the point of intersection from Part A is equivalent to the solution point obtained in Part B</li><li>• <b>Computation component</b> = 1 point: Correct solution of <math>(-2/3, 5/3)</math>.</li></ul> <p>Sample Student Response:</p> $\begin{aligned} \{y=2x+3y=-x+1 \\ 2x+3=-x+1 \\ 3x=-2 \\ x=-2/3 \\ \\ y=-x+1; x=-2/3 \\ y=-(-2/3)+1 \end{aligned}$ |



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | $y = \frac{2}{3}x + 1$<br>$y = \frac{5}{3}$<br>The solution to the system is $(-\frac{2}{3}, \frac{5}{3})$<br>When you graph both equations on a coordinate grid, you plot $y = 2x + 3$ as a line with a slope of 2 and a y-intercept of 3. You plot $y = -x + 1$ as a line with a slope of -1 and a y-intercept of 1. The intersection point is where the two lines meet on the graph. This point satisfies both equations because it lies on both lines. If you substitute the x-coordinate of the intersection point into both equations, you get the same y-coordinate for both.<br><br>By setting the two equations equal to each other, $2x + 3 = -x + 1$ , you are finding the x-coordinate where both lines have the same y-value. Solving this gives the x-coordinate of the intersection point. Substituting the x-value back into either equation gives you the corresponding y-value. $(-\frac{2}{3}, \frac{5}{3})$ is the solution because it is the set of coordinates that satisfies both equations, and that is where the two lines meet on the graph.<br>Or other valid approaches are acceptable. |
| 1 | Student response includes 1 of the 2 elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## #25 Rubric

| Score | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3     | <p>Student response includes the following elements.</p> <ul style="list-style-type: none"><li>• <b>Computation component</b> = 1 point: Identify how much less Atlas Hotel charges per night</li><li>• <b>Modeling component</b> = 1 point: Explain how to determine Atlas Hotel's rate per night</li><li>• <b>Modeling component</b> = 1 point: Explain how to determine Emerald Hotel's rate per night</li></ul> <p>Sample Student Response:</p> <p>The slope of the equation is 175, so Atlas Hotel charges \$175 per night. Emerald Hotel charges <math>\\$540 \div 3</math> nights, or \$180 per night. Atlas Hotel charges less per night. Atlas Hotel charges <math>\\$180 - 175</math>, or \$5 less per night.</p> <p>Or other valid approaches are acceptable.</p> |
| 2     | Student response includes 2 of the 3 elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1     | Student response includes 1 of the 3 elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 0     | Student response is incorrect or irrelevant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |